# KAD Theory: A Control-Theoretic Framework for Alignment Under Partial Observability

**Abstract**

This paper presents Kinetic Alignment of Domains (KAD Theory), a unified control-theoretic framework for modeling failure across engineered, cognitive, and organizational systems. The theory formalizes misalignment as the fundamental driver of failure under partial and dynamic observability. By embedding agent–domain interaction within a measurable Hilbert space and introducing visibility as a conditional expectation operator, KAD establishes a closed dynamical system governing alignment. A key result is the Information–Failure Law, which proves that unobserved domain components impose a strict lower bound on achievable alignment. The framework further introduces dual control over action and perception, yielding a generalized stability criterion under stochastic drift. This work provides a mathematically rigorous foundation for alignment, bridging control theory, AI safety, and socio-technical governance.

## 1 Introduction

Failure across engineered, cognitive, and organizational systems exhibits striking structural regularities despite substantial differences in scale, mechanism, and domain. Traditional theories characterize failure in terms of stochastic degradation, instability, miscalibration, or human error, yet these explanations remain fragmented and domain-specific. What is missing is a unified mathematical account that explains *why* systems fail in the presence of uncertainty, adaptation, and incomplete information.

KAD Theory proposes that all such failures share a single underlying mechanism: *misalignment between an agent and its operational domain under partial observability.* Rather than treating failure as a breakdown of components or policies, KAD frames it as a dynamical divergence between what the system is optimizing and what the environment requires it to optimize, given the information available to it.

The central hypothesis of KAD Theory is:

> **Perfect alignment is impossible without full visibility of the operational domain.**

This insight reframes alignment as a problem of *control under information constraints*, integrating perception, learning, adaptation, and governance into a single measurable dynamical system. By doing so, KAD Theory provides a unified mathematical structure capable of describing failure across artificial intelligence, engineered control systems, human cognition, and organizational decision-making.

## 2 Mathematical Foundations

### 2.1 State Space

We model both the domain and the agent as elements of a common functional space. Let

$$(D(t), A(t)) \in \mathcal{H}, \qquad \mathcal{H} = L^2(X, \mu),$$

where $X$ is the underlying domain of semantic features and $\mu$ is a reference measure. Here,

- $D(t)$ denotes the domain state (environment, task semantics, latent requirements),
- $A(t)$ denotes the agent state (policy, cognitive model, or strategy).

Embedding both processes in $L^2$ provides a metric and geometric structure suitable for orthogonal projections, conditional expectations, and Lyapunov analysis, enabling a unified treatment of perception, adaptation, and misalignment.

### 2.2 Visibility Operator

A central construct in KAD Theory is the visibility operator, defined through the observable information structure. Let $\mathcal{F}_{\text{obs}}(t)$ denote the $\sigma$-algebra generated by all information available to the agent at time $t$. The visibility operator is the conditional expectation

$$P = \mathbb{E}[\cdot \mid \mathcal{F}_{\text{obs}}(t)],$$

which acts as an orthogonal projection onto the subspace of observable signals. Thus, $P$ satisfies $P^2 = P$ and is self-adjoint on $\mathcal{H}$.

The unobservable or "dark" component of the domain is

$$D_{\text{dark}} = (I - P)D,$$

representing latent structure that cannot be inferred or acted upon by the agent. This term plays a central role in establishing fundamental limits on achievable alignment.

### 2.3 Alignment Error

Alignment is defined as the discrepancy between the agent's representation and the domain's requirements restricted to the observable subspace:

$$E(t) = |PD(t) - PA(t)|.$$

This formulation captures the fact that alignment is inherently information-bounded: only the visible components of the domain can be aligned. Misalignment arising from $(I-P)D$ is irreducible, forming the basis of the Information–Failure Law. Alignment is achieved when $E(t) \to 0$, a limit that is unattainable unless the dark component vanishes.

## 3 System Dynamics

The evolution of alignment in KAD Theory is obtained by differentiating the projected domain–agent discrepancy. Let

$$E(t) = PD(t) - PA(t).$$

Applying the product rule for operator-valued functions yields the alignment dynamics

$$\dot{E}(t) = P(W - U) + \lambda PA + \dot{P}(D - A),$$

where:

- $W$ denotes domain drift (exogenous stochastic change),
- $U$ is the agent's adaptation or learning policy,
- $\lambda$ is the cognitive decay or focus-loss rate,
- $\dot{P}$ captures the dynamics of the visibility operator.

Each term corresponds to a distinct mechanism influencing alignment:

- $PW$ represents the visible component of domain drift, the primary driver of misalignment growth.
- $-PU$ quantifies corrective adaptation within the observable subspace.
- $\lambda PA$ models degradation of the agent's internal representation.
- $\dot{P}(D - A)$ captures changes in perceptual bandwidth, revealing or concealing latent misalignment.

This equation provides a unified dynamical law coupling drift, learning, cognition, and perception, forming the basis for the Information–Failure Law and the stability results developed in subsequent sections.

## 4 Information Failure Law

The Information–Failure Law formalizes the central insight of KAD Theory: *misalignment cannot be reduced below the magnitude of the unobservable domain.* This result is not merely intuitive—it follows directly from the geometry of conditional expectation in Hilbert spaces. Because the visibility operator $P$ is an orthogonal projection, the dark component $(I - P)D$ lies in a subspace that is, by definition, inaccessible to the agent. No learning rule, control input, or governance mechanism can compensate for information that is not observable.

[Information–Failure Law: Irreducible Lower Bound] Let $D(t), A(t) \in \mathcal{H}$ and let

$$P = \mathbb{E}[\cdot \mid \mathcal{F}_{\text{obs}}(t)]$$

be the orthogonal projection onto the observable subspace. Then for all $t$,

$$\|E(t)\| = \|PD(t) - PA(t)\| \quad \geq \quad c\,\|(I - P)D(t)\|,$$

for some constant $c > 0$ determined by the geometry of the projection and the norm on $\mathcal{H}$.
**Interpretation.** The term $(I-P)D(t)$ represents the *dark domain*—the portion of the environment that the agent cannot perceive or infer. The inequality states that alignment error cannot be made arbitrarily small unless the dark domain itself is zero. Even perfect adaptation ($A(t) = D(t)$) cannot eliminate misalignment if visibility is incomplete. This establishes a fundamental limit on alignment under partial observability.

**Perfect alignment requires full observability.**

No control law, learning algorithm, or governance mechanism can overcome the geometric constraint imposed by the projection operator. Misalignment is therefore not only a control problem—it is an *information-theoretic inevitability*.

# 5 Stability Analysis

Stability in KAD Theory concerns whether the alignment error $E(t)$ remains bounded despite drift, cognitive decay, and partial observability. Because the system evolves in a Hilbert space and the visibility operator $P$ is time-varying, classical linear stability tools are insufficient. Instead, we employ a Lyapunov functional that captures both geometric and information-theoretic structure.

## 1. Lyapunov Function

We consider the natural quadratic Lyapunov function

$$V(E) = \|E(t)\|^2,$$

which measures the squared misalignment within the observable subspace. Differentiating along system trajectories yields

$$\dot{V}(t) = 2\langle E, \dot{E}\rangle.$$

Substituting the alignment dynamics and applying Cauchy–Schwarz and projection inequalities gives

$$\dot{V}(t) \;\leq\; -c\|E(t)\|^2 \;+\; \|(I-P)D(t)\|^2,$$

for some constant $c > 0$ determined by the adaptation rate and the geometry of $P$.
This inequality reveals two competing forces:

- a negative term representing alignment correction through adaptation and stability of the observable subspace,
- a positive term representing irreducible misalignment injected by the dark domain.

## 2. Uniform Ultimate Boundedness

The inequality above has the standard form of an input-to-state stability (ISS) condition, where the "input" is the unobservable domain component $(I-P)D$. This leads directly to the following result.
[Uniform Boundedness of Alignment Error] If:

- adaptation dominates drift, i.e.

$$\langle E, P(W - U)\rangle \leq -c\|E\|^2 \quad \text{for some } c > 0,$$

- dark information is bounded, i.e.

$$\|(I - P)D(t)\| \leq \delta < \infty,$$

then the alignment error $E(t)$ is uniformly ultimately bounded. Specifically,

$$\|E(t)\| \;\leq\; \frac{1}{\sqrt{c}}\,\delta \quad \text{for all sufficiently large } t.$$

Thus the system converges to a neighborhood whose radius is proportional to the magnitude of the dark domain.

**Interpretation.** If the agent adapts faster than the domain drifts, alignment error decays. But because the dark domain acts as a persistent disturbance, error cannot converge to zero. The best achievable alignment is a ball whose size is determined by $\|(I - P)D\|$.

This result formalizes the fundamental limit: alignment is stable but never perfect under partial observability.

sectionDual Control Problem

Classical control theory assumes that the controller has fixed access to observations. KAD Theory breaks this assumption by treating perception itself as a controllable resource. The agent must therefore solve a dual optimization problem involving both action and perception.

## Action and Perception as Joint Decisions

The agent must simultaneously determine:

- **Action control**: how to adapt the internal state $A(t)$ to reduce misalignment,
- **Perception control**: how to shape the visibility operator $P(t)$ to acquire information about the domain.

This yields a fundamentally richer structure than standard adaptive control or reinforcement learning, because the agent must jointly decide:

- what to do, and
- what to look at.

## Dual Optimization Formulation

The agent selects both an adaptation policy $U(t)$ and a visibility policy $P(t)$ by solving

$$(U, P) = \arg\min_{U,\,P}\; \mathbb{E}\big[\|E(t)\|^2 \;+\; \gamma\, C(P)\big]\,,$$

where:

- $\|E(t)\|^2$ penalizes misalignment in the observable subspace,

- $C(P)$ is the cost of acquiring or expanding visibility (e.g., sensing, computation, interpretability, governance),
- $\gamma > 0$ trades off alignment accuracy against information cost.

This formulation captures the essential tension:

- more visibility reduces misalignment,
- but increases sensing cost, cognitive load, or governance overhead.

### Interpretation

The optimization produces two coupled outputs:

1. **Action policy** $U^*(t)$: determines how the agent adapts its internal representation to track the domain. This is the classical control component.
2. **Sensing policy** $P^*(t)$: determines what the agent chooses to observe, infer, or monitor. This is the novel component introduced by KAD Theory.

The sensing policy may:

- expand the observable subspace,
- reallocate attention,
- increase interpretability,
- request additional governance-level visibility.

Thus, KAD Theory treats perception as a *strategic decision*, not a passive constraint.

### Significance

This dual-control formulation is the mathematical foundation for:

- adaptive interpretability,
- dynamic oversight,
- attention allocation,
- governance-aware sensing,
- and the Information–Failure Law.

It shows that alignment is not only about acting correctly but also about *seeing correctly*.

# 6 Phase Transition Criterion

The alignment dynamics of KAD Theory exhibit a sharp phase transition governed by the relative strength of adaptation versus domain drift. This transition is captured by the dimensionless stability ratio

$$\rho = \frac{\|PU\|}{\|PW\|},$$

which compares the effective corrective power of the agent (projected adaptation) with the effective disturbance power of the environment (projected drift). Because both terms are projected through the visibility operator $P$, the ratio measures stability within the observable subspace.

### Interpretation of the Stability Ratio

**1. $\rho > 1$: Alignment-Dominant Regime (Stable Phase).** When projected adaptation exceeds projected drift, the agent can counteract changes in the domain faster than they accumulate. Alignment error decays, the Lyapunov derivative becomes negative, and the system enters a stable basin determined by the dark domain.

**2. $\rho < 1$: Drift-Dominant Regime (Divergent Phase).** When drift overwhelms adaptation, the agent cannot keep pace with environmental change. Misalignment grows, the Lyapunov derivative becomes positive, and the system diverges regardless of control effort.

### Phase Transition Interpretation

The ratio $\rho$ induces a nonlinear phase transition analogous to stability thresholds in dynamical systems and critical points in statistical physics:

$$\rho > 1 \Rightarrow \text{stable alignment}, \qquad \rho < 1 \Rightarrow \text{divergence}.$$

The boundary $\rho = 1$ marks the critical threshold separating stable and unstable alignment dynamics. Small changes in visibility, adaptation, or drift can push the system across this boundary, making the phase transition a central object for governance and control.

# 7 Safety and Invariance

Safety in KAD Theory concerns ensuring that the alignment error $E(t)$ remains within an acceptable region despite drift, partial observability, and cognitive decay. Because misalignment evolves under both controllable and uncontrollable influences, safety must be enforced through forward invariance of a prescribed safe set.

### 1. Safety Set

We define the admissible alignment region as

$$\mathcal{S} = \{E : \|E\| \leq \epsilon\},$$

where $\epsilon > 0$ specifies the maximum tolerable misalignment. This set represents the operational safety envelope of the system.

### 2. Barrier Function

To ensure that trajectories remain inside $\mathcal{S}$, we introduce the barrier function

$$h(E) = \epsilon^2 - \|E\|^2.$$

The condition $h(E) \geq 0$ is equivalent to $E \in \mathcal{S}$, so maintaining safety is equivalent to preserving non-negativity of the barrier.

### 3. Invariance Condition

Forward invariance of $\mathcal{S}$ is guaranteed if the barrier satisfies the exponential condition

$$\dot{h}(t) \geq -\kappa h(t), \qquad \kappa > 0,$$

which is the standard control barrier condition. This inequality ensures that:

- if the system is inside the safe set, it remains inside,
- if it approaches the boundary, corrective action is enforced,
- violations are exponentially discouraged.

Substituting $h(E)$ yields

$$\dot{h}(t) = -2\langle E, \dot{E}\rangle \geq -\kappa(\epsilon^2 - \|E\|^2).$$

This condition couples safety directly to the alignment dynamics, making safety a structural property of the KAD system rather than an external constraint.

### Interpretation

The safe set $\mathcal{S}$ defines the region of acceptable misalignment. The barrier condition ensures that the system cannot exit $\mathcal{S}$ unless the dark domain or drift overwhelms adaptation. Governance or oversight mechanisms can be interpreted as modifying $P$, $U$, or $\kappa$ to maintain invariance.

Thus, KAD Theory embeds safety directly into the geometry of alignment, providing a principled method for ensuring safe operation under partial observability.

## 8 Implications

KAD Theory provides a unifying mathematical lens for understanding failure across diverse domains. Because misalignment under partial observability is a structural phenomenon, the theory yields deep implications for engineered systems, artificial intelligence, human cognition, and organizational behavior.

### 8.1 Engineering Systems

In engineered control systems, failure often arises from unmodeled dynamics, sensor limitations, and latent disturbances. KAD Theory formalizes these issues as dark-domain components $(I-P)D$ that cannot be sensed, insufficient corrective authority when $\rho < 1$, and loss of invariance when drift overwhelms adaptation. This reframes robustness and fault tolerance as information-bounded alignment problems.

### 8.2 Artificial Intelligence

In AI systems, KAD Theory maps directly onto distribution shift (domain drift $W$), hidden capabilities (dark domain $(I - P)D$), interpretability and oversight limits (visibility operator $P$), and policy drift or catastrophic forgetting (cognitive decay $\lambda A$). The Information–Failure Law shows that misalignment is irreducible unless visibility is expanded, providing a mathematical foundation for interpretability, monitoring, and governance.

### 8.3 Human Cognition

Human cognitive failure—attention lapses, situational awareness loss, and misjudgment—emerges naturally from the KAD framework. Attention limits correspond to a restricted visibility operator $P$, cognitive decay models forgetting and focus loss, and bounded rationality arises from the cost of expanding perception $C(P)$. KAD thus treats perception and attention as strategic, resource-bounded processes.

### 8.4 Organizations

Organizational failure is reinterpreted as strategic misalignment under incomplete information. Different units operate with distinct visibility operators $P_i$, governance defines the admissible region $\Omega$, and coordination failures arise from misaligned projections of the same domain. KAD provides a mathematical foundation for organizational design and governance by showing how information asymmetry and drift drive systemic misalignment.

## 9 Conclusion

KAD Theory provides a unified mathematical foundation for understanding alignment, failure, and stability across engineered, cognitive, artificial, and organizational systems. By embedding agent–domain interaction within a Hilbert-space geometry and introducing visibility as an operator-level constraint, the framework reframes alignment as an information-bounded control problem.

KAD Theory establishes:

- alignment as a measurable dynamical variable governed by projection geometry,
- visibility as the structural determinant of knowledge and inference,
- failure as the inevitable consequence of irreducible dark-domain components.

These results show that misalignment is not merely a failure of optimization or control, but a structural property of systems operating under partial observability.

> **Alignment is not only a control problem—it is fundamentally an information problem.**

KAD Theory thus offers a principled mathematical basis for designing adaptive controllers, interpretable AI systems, cognitive architectures, and governance mechanisms capable of operating safely in dynamic and uncertain environments.

## 10 Future Work

KAD Theory opens several mathematically rich and practically consequential research directions. Because the framework unifies control, perception, learning, and governance under partial observability, its extensions naturally span nonlinear systems, high-dimensional inference, empirical validation, and real-world deployment.

- **Nonlinear extensions:** Extending KAD dynamics beyond linear Hilbert-space projections to nonlinear operators, nonlinear visibility maps, and nonlinear drift processes, including contraction analysis and differential geometric formulations.
- **High-dimensional domain modeling:** Developing scalable representations of $D(t)$ and $A(t)$ for complex environments such as multimodal AI systems, cyber-physical infrastructures, and socio-technical networks.
- **Empirical validation of the Information–Failure Law:** Testing the predicted lower bound on alignment error across engineered systems, machine learning models, human cognition, and organizational decision processes.
- **Real-time estimation of visibility operators:** Designing algorithms to infer or approximate the visibility operator $P(t)$ online, including adaptive sensing, attention allocation, interpretability mechanisms, and governance-driven oversight.
- **Applications in autonomous and organizational systems:** Applying KAD Theory to autonomous vehicles, robotics, large-scale AI systems, human-machine teams, and organizational governance structures through dual-control architectures that jointly optimize action and perception.

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# Appendix

## A Tables

Table 1: Summary of Core Operators in KAD Theory

| **Symbol** | **Name** | **Interpretation** |
|---|---|---|
| $D(t)$ | Domain State | True environment or task structure |
| $A(t)$ | Agent State | Internal model or learned representation |
| $P(t)$ | Visibility Operator | Observable subspace / information access |
| $W(t)$ | Domain Drift | Exogenous change in the environment |
| $U(t)$ | Adaptation Policy | Agent's corrective action |

Table 2: Stability and Failure Conditions

| **Condition** | **Criterion** | **Outcome** |
|---|---|---|
| $\rho > 1$ | Adaptation dominates drift | Stable alignment |
| $\rho < 1$ | Drift dominates adaptation | Divergence |
| $\|(I - P)D\| = 0$ | Full observability | Perfect alignment possible |
| $\|(I - P)D\| > 0$ | Partial observability | Irreducible error floor |

# B Additional Sections

## B.1 Mathematical Proofs

This appendix provides detailed proofs of the core theoretical results introduced in the main text, including the Information–Failure Law, Lyapunov stability guarantees, and the barrier invariance condition for safe operation.

### B.1.1 Proof of the Information–Failure Law

**Theorem.** For any visibility operator $P$ and domain state $D(t)$,

$$\|E(t)\| = \|PD(t) - PA(t)\| \;\geq\; c\,\|(I-P)D(t)\|,$$

for some constant $c > 0$ determined by the geometry of the projection.

**Proof sketch.** Because $P$ is an orthogonal projection, the Hilbert space decomposes as

$$\mathcal{H} = \mathrm{Range}(P) \oplus \mathrm{Null}(P).$$

The dark-domain component $(I-P)D(t)$ lies entirely in $\mathrm{Null}(P)$ and is orthogonal to all observable components. Any estimator $A(t)$ must satisfy

$$PA(t) \in \mathrm{Range}(P),$$

and therefore cannot reduce the norm of $(I-P)D(t)$. The inequality follows from the Pythagorean decomposition and the fact that the projection error cannot be smaller than the magnitude of the unobservable component. □

### B.1.2 Lyapunov Stability Proof

Let the Lyapunov function be $V(E) = \|E\|^2$. Differentiating along system trajectories gives

$$\dot{V}(t) = 2\langle E, \dot{E}\rangle.$$

Under the condition that projected adaptation dominates projected drift,

$$\|PU\| > \|PW\|,$$

the inner product becomes negative definite, ensuring

$$\dot{V}(t) < 0,$$

and therefore exponential convergence of $E(t)$ to a bounded region determined by the dark domain. □

### B.1.3 Barrier Invariance Proof

Define the barrier function

$$h(E) = \epsilon^2 - \|E\|^2.$$

Forward invariance of the safe set $\mathcal{S} = \{E : \|E\| \leq \epsilon\}$ is guaranteed if

$$\dot{h}(t) \geq -\kappa h(t), \qquad \kappa > 0.$$

Differentiating,

$$\dot{h}(t) = -2\langle E, \dot{E} \rangle.$$

Substituting the system dynamics and applying the exponential barrier condition yields

$$-2\langle E, \dot{E} \rangle \geq -\kappa(\epsilon^2 - \|E\|^2),$$

which ensures that trajectories cannot exit $\mathcal{S}$ unless drift or dark-domain components overwhelm adaptation. □

## B.2 Extended Derivations

This section provides extended derivations for the core mathematical components of KAD Theory, including the alignment dynamics, projection geometry, and the dual-control optimization framework.

### B.2.1 Derivation of Alignment Dynamics

Starting from the definition of alignment error

$$E(t) = P(t)D(t) - P(t)A(t),$$

we differentiate with respect to time:

$$\dot{E}(t) = \dot{P}(t)D(t) + P(t)\dot{D}(t) - \dot{P}(t)A(t) - P(t)\dot{A}(t).$$

Grouping terms yields

$$\dot{E}(t) = P(t)\big(\dot{D}(t) - \dot{A}(t)\big) + \dot{P}(t)\big(D(t) - A(t)\big).$$

Identifying domain drift as $W(t) = \dot{D}(t)$ and adaptation as $U(t) = \dot{A}(t)$ gives the alignment dynamics:

$$\dot{E}(t) = P(t)\big(W(t) - U(t)\big) + \dot{P}(t)\big(D(t) - A(t)\big).$$

The additional cognitive decay term $\lambda P(t)A(t)$ arises from modeling focus loss as a first-order leakage process.

### B.2.2 Projection Geometry

The visibility operator $P$ is modeled as an orthogonal projection on a Hilbert space $\mathcal{H}$, satisfying

$$P^2 = P, \qquad P^* = P.$$

This induces the decomposition

$$\mathcal{H} = \mathrm{Range}(P) \oplus \mathrm{Null}(P),$$

corresponding to observable and unobservable components. Any domain state $D$ decomposes as

$$D = PD + (I - P)D,$$

where $(I - P)D$ represents the dark-domain component that cannot be inferred by the agent. This geometric structure underlies the Information–Failure Law and the irreducible error floor.

### B.2.3 Dual-Control Optimization

The agent jointly selects an adaptation policy $U(t)$ and a visibility policy $P(t)$ by solving

$$(U^*, P^*) = \arg\min_{U,\,P} \mathbb{E}\big[\|E(t)\|^2 + \gamma\, C(P)\big]\,.$$

Expanding the objective using the alignment dynamics yields

$$\|E(t)\|^2 = \|P(D - A)\|^2,$$

which depends explicitly on the choice of $P$. The cost term $C(P)$ penalizes expansion of the observable subspace, capturing sensing, computation, interpretability, and governance overhead.

Taking functional derivatives with respect to $U$ and $P$ gives the coupled optimality conditions:

$$\frac{\delta}{\delta U}\mathbb{E}\|E(t)\|^2 = 0, \qquad \frac{\delta}{\delta P}\mathbb{E}\big[\|E(t)\|^2 + \gamma C(P)\big] = 0.$$

These yield:

- an optimal adaptation policy $U^*(t)$ that minimizes projected misalignment,
- an optimal sensing policy $P^*(t)$ that balances information gain against cost.

The resulting structure generalizes classical dual control by treating perception as a controllable resource rather than a fixed constraint.

## B.3 Experimental Setup

If empirical validation is performed, this section documents the datasets, metrics, and evaluation protocols used to test the predictions of KAD Theory. The goal is to measure alignment error, visibility dynamics, and the stability ratio $\rho$ under controlled and real-world conditions.

### B.3.1 Datasets

Experiments may draw from:

- **Synthetic environments:** controlled simulations of domain drift $W(t)$, visibility constraints $P(t)$, and adaptation policies $U(t)$.
- **Machine learning benchmarks:** datasets exhibiting distribution shift, such as time-varying classification or reinforcement learning tasks.
- **Human-in-the-loop tasks:** attention-limited decision-making scenarios for validating cognitive decay and partial observability effects.
- **Organizational datasets:** multi-agent decision logs with heterogeneous visibility operators $P_i$.

### B.3.2 Metrics

Key metrics include:

- **Alignment error:** $\|E(t)\| = \|P(D - A)\|$ measured over time.
- **Dark-domain magnitude:** $\|(I - P)D(t)\|$ as a predictor of irreducible error.
- **Stability ratio:** $\rho = \|PU\|/\|PW\|$ to identify phase transitions.
- **Visibility cost:** $C(P)$ quantifying sensing, computation, or governance overhead.

### B.3.3 Evaluation Protocol

The evaluation protocol typically includes:

1. Initializing domain and agent states $(D_0, A_0)$ under a chosen visibility operator $P_0$.
2. Introducing controlled drift $W(t)$ or real-world distribution shift.
3. Applying an adaptation policy $U(t)$ or learned controller.
4. Measuring alignment error, stability ratio, and barrier satisfaction over time.
5. Comparing empirical results to theoretical predictions of the Information–Failure Law.

### B.3.4 Reproducibility

All experiments should specify:

- random seeds and initialization procedures,
- simulation time horizons and step sizes,
- hardware and software configurations,
- hyperparameters for adaptation or learning algorithms.

This ensures that empirical validation of KAD Theory is transparent, repeatable, and comparable across different domains and research groups.

## B.4 Implementation Notes

This section summarizes practical considerations for implementing simulations, numerical experiments, and operator-based computations used in validating KAD Theory.

### B.4.1 Numerical Solvers

The alignment dynamics

$$\dot{E}(t) = P(t)\big(W(t) - U(t)\big) + \dot{P}(t)\big(D(t) - A(t)\big)$$

are typically integrated using standard ODE solvers. Common choices include:

- explicit Euler for fast prototyping,

- Runge–Kutta (RK4) for stable medium-accuracy integration,
- adaptive solvers (e.g., Dormand–Prince) for stiff or rapidly varying drift processes.

Stability is sensitive to the time step $\Delta t$, especially when modeling rapid domain drift or high-gain adaptation.

#### B.4.2 Operator Discretization

The visibility operator $P(t)$ and its complement $(I - P(t))$ are implemented as:

- projection matrices in finite-dimensional simulations,
- orthogonal basis masks for high-dimensional embeddings,
- kernel-induced projections for functional or infinite-dimensional settings.

Time-varying visibility is discretized as $P_k = P(t_k)$, with $\dot{P}(t)$ approximated by finite differences:

$$\dot{P}(t_k) \approx \frac{P_{k+1} - P_k}{\Delta t}.$$

#### B.4.3 Simulation Environment

Experiments typically include:

- **Domain drift models:** stochastic processes (e.g., Gaussian drift), adversarial perturbations, or structured regime shifts.
- **Adaptation policies:** gradient-based updates, control laws, or learned policies.
- **Visibility schedules:** fixed projections, periodic sensing, or adaptive attention mechanisms.

Simulation horizons, noise levels, and drift magnitudes are selected to test stability, invariance, and the Information–Failure Law under diverse conditions.

#### B.4.4 Computational Considerations

Efficient implementation requires:

- vectorized operations for large-scale projections,
- caching of repeated operator evaluations,
- GPU acceleration for high-dimensional embeddings,
- reproducible random seeds and logging of all hyperparameters.

These practices ensure that numerical results are stable, interpretable, and replicable across different computational environments.